

Java Important Topics (Basic to Advanced)

Java Important Topics (Basic → Advanced)

1. Basics of Java

- Variables
- Data Types
- Operators
- Type Casting
- Input/Output (Scanner)
- Keywords
- Comments

2. Control Statements

- if
- if-else
- switch
- Loops: for, while, do-while
- break
- continue

3. Methods

- Method Creation
- Parameters
- Return Type
- Method Overloading
- Varargs

4. Arrays

- 1D Array
- 2D Array
- Enhanced for loop
- Arrays.sort()
- Arrays.toString()
- Arrays.deepToString()
- Arrays.parallelSort()

5. String

- String Methods
- String Comparison
- Immutable Nature
- StringBuilder
- StringBuffer

6. Object Oriented Programming

- Class & Object
- Constructor
- this
- static
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Interface

7. Important Object Class Methods

- toString()
- equals()
- hashCode()

8. Packages & Access Modifiers

- Packages
- public
- private
- protected
- default

9. Exception Handling

- try
- catch
- finally
- throw
- throws
- Custom Exception

10. Collections Framework

List:

- ArrayList
- LinkedList
- Vector
- Stack

Set:

- HashSet
- LinkedHashSet
- TreeSet

Queue:

- PriorityQueue
- Deque

Map:

- HashMap
- LinkedHashMap
- TreeMap

11. Sorting in Java

- Arrays.sort()
- Collections.sort()
- Comparable
- Comparator
- list.sort()
- Stream sorting

12. Generics

- Generic Class
- Generic Method
- Wildcards

13. File Handling

- File Class
- FileReader
- FileWriter
- BufferedReader
- BufferedWriter

14. Multithreading

- Thread Class
- Runnable Interface
- Synchronization
- Thread Lifecycle

15. Java 8 Features

- Lambda Expression
- Functional Interface
- Stream API
- Method Reference
- Optional

16. Stream API

- filter()
- map()
- sorted()
- collect()
- forEach()
- reduce()

17. JDBC

- Database Connection
- PreparedStatement
- ResultSet

18. Serialization

- Serializable Interface
- Object Streams

19. Important Utility Classes

- Math
- Arrays
- Collections
- Objects
- Scanner

20. Advanced Java

- Reflection API
- Annotations

- Executor Framework
- Concurrency
- CompletableFuture

Most Important for Interview:

- OOPs
- String
- Collections
- Exception Handling
- Multithreading
- Comparable vs Comparator
- HashMap Internal
- equals() vs ==
- final, finally, finalize

Most Important for Development:

- Collections
- Stream API
- Lambda
- Generics
- File Handling
- Multithreading
- JDBC